

IMAT MOCK EXAM 2 — "Bioenergetics Spine"

60 questions · 100 minutes · scoring: +1.5 correct, -0.4 wrong, 0 blank · personal target: 47/90.0 · Mock exam

Section A - Logical Reasoning & Reading (Q1-9)

1. "Antibiotic resistance arises when bacteria acquire genetic changes — by mutation or horizontal gene transfer — that reduce susceptibility to a drug. Resistant strains survive treatment and proliferate, so each course of antibiotics applies selective pressure favouring resistance. Resistance spreads faster in populations with high antibiotic use."

Which is the best-supported conclusion?

- A) Reducing unnecessary antibiotic use would slow the spread of resistance.
- B) Antibiotic use creates genetic mutations directly.
- C) All bacteria become resistant after one course.
- D) Resistance is independent of selection pressure.
- E) Resistance cannot arise without horizontal gene transfer.

2. "Homeostasis refers to the maintenance of a constant internal environment despite external changes. Core variables such as body temperature, blood pH, and blood glucose are regulated by negative feedback loops: a sensor detects deviation, a control centre orchestrates a response, and an effector returns the variable to its set point."

Which is consistent with the passage?

- A) Homeostasis requires no sensor.
- B) Negative feedback returns a variable toward its set point.
- C) A positive-feedback loop is the main homeostatic mechanism.
- D) Body temperature is not a homeostatically regulated variable.
- E) The effector's job is to push the variable further from the set point.

3. (General knowledge) The Italian Renaissance anatomist who published *De humani corporis fabrica* (1543), correcting many Galenic errors through direct dissection, was:

- A) Leonardo da Vinci
- B) Marcello Malpighi
- C) Andreas Vesalius
- D) Luigi Galvani
- E) William Harvey

4. Four students — P, Q, R, S — are ranked in an exam. P scored higher than Q but lower than R. S scored lower than Q. Who scored lowest?

- A) P
- B) R
- C) Q
- D) Cannot be determined
- E) S

5. "If it rains, the match is cancelled." The match was not cancelled. Therefore:

- A) The teams refused to play.
- B) It rained.
- C) The match was cancelled.
- D) It did not rain.
- E) It might have rained.

6. A machine applies the rule: take a number, square it, then subtract 4. If the output is 32, the input could be:

- A) 4 B) 5 C) 6 D) 7 E) 8

7. At 9:45, what is the angle between the hour and minute hands of a clock?

- A)** 0° **B)** 15° **C)** 22.5° **D)** 30° **E)** 45°

8. A is the father of B. B is the sister of C. How is A related to C?

- A)** Grandfather **B)** Uncle **C)** Cousin **D)** Brother **E)** Father

9. Complete the series: 1, 4, 9, 16, 25, ?

- A)** 30 **B)** 33 **C)** 36 **D)** 49 **E)** 64

Section B - Biology (Q10-32)

10. Glycolysis occurs in the:

- A)** Nucleus
B) Inner mitochondrial membrane
C) Mitochondrial matrix
D) Cytosol
E) Chloroplast stroma

11. Under anaerobic conditions in human muscle, pyruvate is converted to:

- A)** Lactate (lactic acid)
B) Acetate
C) Acetyl-CoA
D) Glucose
E) Ethanol and CO₂

12. The net reaction of aerobic respiration (per glucose) produces approximately how many ATP in eukaryotes (modern estimate)?

- A)** 2 **B)** 4 **C)** 6-8 **D)** 30-32 **E)** 100

13. Which coenzyme carries electrons from glycolysis and the Krebs cycle to the electron transport chain?

- A)** Coenzyme A
B) NAD⁺ / NADH and FAD / FADH₂
C) ADP
D) ATP
E) AMP

14. The proton-motive force across the inner mitochondrial membrane is generated by:

- A)** Pumping of H⁺ from matrix to intermembrane space by electron transport complexes
B) Pyruvate transport
C) Diffusion of O₂
D) Ca²⁺ release
E) ATP hydrolysis

15. Which metabolic pathway occurs in both cellular respiration and photosynthesis?

- A)** Calvin cycle
B) The Krebs cycle
C) Light-dependent reactions
D) Glycolysis-like reactions of the Calvin cycle only
E) Electron transport chain with chemiosmosis

16. In which organelle does photosynthesis take place in plant cells?

- A)** Peroxisome **B)** Chloroplast **C)** Mitochondrion **D)** Nucleus **E)** Vacuole

- 17.** The light-dependent reactions of photosynthesis produce:
- A)** ATP, NADPH, and O₂
 - B)** Glucose and O₂
 - C)** Pyruvate
 - D)** Lactate
 - E)** CO₂ and H₂O
- 18.** Which blood vessels carry oxygenated blood from the lungs to the heart?
- A)** Coronary arteries
 - B)** Pulmonary veins
 - C)** Vena cavae
 - D)** Pulmonary arteries
 - E)** Aorta
- 19.** Gas exchange in the lungs occurs across the walls of the:
- A)** Bronchi **B)** Trachea **C)** Aorta **D)** Alveoli **E)** Pharynx
- 20.** Which cells of the gastric glands secrete hydrochloric acid?
- A)** Enterocytes
 - B)** Mucous neck cells
 - C)** Goblet cells
 - D)** Chief cells
 - E)** Parietal (oxyntic) cells
- 21.** Bile, which emulsifies fats, is produced by the __ and stored in the __.
- A)** Gallbladder / liver
 - B)** Pancreas / gallbladder
 - C)** Stomach / pancreas
 - D)** Liver / gallbladder
 - E)** Small intestine / liver
- 22.** The basic functional unit of the kidney is the:
- A)** Sarcomere **B)** Alveolus **C)** Nephron **D)** Hepatocyte **E)** Neuron
- 23.** Filtration of blood in the nephron occurs at the:
- A)** Collecting duct
 - B)** Glomerulus (Bowman's capsule)
 - C)** Loop of Henle
 - D)** Renal pelvis
 - E)** Distal convoluted tubule
- 24.** Which tissue lines the inner surface of the stomach and intestines?
- A)** Simple columnar epithelium
 - B)** Simple cuboidal epithelium
 - C)** Stratified squamous epithelium
 - D)** Pseudostratified ciliated epithelium
 - E)** Transitional epithelium
- 25.** Smooth muscle is found in the walls of all of the following EXCEPT:
- A)** The uterus
 - B)** The bronchi
 - C)** Blood vessels
 - D)** The intestines
 - E)** Skeletal attachments to bone (tendons are connective tissue, not muscle)

26. Which connective tissue cell type produces the fibres of regular dense connective tissue (e.g., tendons)?

- A)** Erythrocytes **B)** Fibroblasts **C)** Osteocytes **D)** Chondrocytes **E)** Adipocytes

27. The primary stimulus for ventilation during moderate exercise is the rise in:

- A)** Blood O_2
B) Blood nitrogen
C) Atmospheric pressure
D) Blood CO_2 (and H^+), sensed by central and peripheral chemoreceptors
E) Body temperature only

28. Insulin promotes which of the following in liver and muscle?

- A)** Glycogenolysis and gluconeogenesis
B) Lipolysis
C) Glycogen synthesis (glycogenesis) and glucose uptake
D) Ketogenesis
E) Glucose release into blood

29. The long bone of the thigh is the:

- A)** Tibia **B)** Fibula **C)** Humerus **D)** Patella **E)** Femur

30. A red blood cell placed in a hypertonic solution will:

- A)** Shrink (crenate)
B) Divide
C) Remain unchanged
D) Swell and burst
E) Become a leucocyte

31. The phase of mitosis in which chromosomes align at the cell's equator is:

- A)** Anaphase **B)** Metaphase **C)** Telophase **D)** Cytokinesis **E)** Prophase

32. Which of the following is the correct base-pairing rule in DNA?

- A)** A – U and G – C (this is DNA-RNA, not DNA-DNA)
B) A – C and G – T
C) A – G and T – C
D) A – T and G – C
E) A – A and T – T

Section C - Chemistry (Q33-47)

33. Balance: $__ Al + __ O_2 \rightarrow __ Al_2O_3$. The coefficient of O_2 is:

- A)** 1 **B)** 2 **C)** 3 **D)** 4 **E)** 6

34. What is the oxidation number of Mn in $KMnO_4$?

- A)** –1 **B)** +2 **C)** +4 **D)** +6 **E)** +7

35. 25.0 mL of 0.10 M HCl is neutralised by 0.05 M NaOH. What volume of NaOH is required?

- A)** 5 mL **B)** 12.5 mL **C)** 25 mL **D)** 50 mL **E)** 100 mL

36. A buffer solution resists changes in pH when small amounts of acid or base are added. A typical buffer consists of:

- A)** A weak acid and its conjugate base (or weak base and its conjugate acid)
B) A strong acid and a strong base
C) Pure water
D) Two strong acids
E) A strong acid and its conjugate base

37. Which has the highest electronegativity?
A) F B) Cs C) Na D) Cl E) O
38. The shape of a methane (CH_4) molecule is:
**A) Linear
B) Tetrahedral
C) Octahedral
D) Pyramidal
E) Trigonal planar**
39. At STP, what volume is occupied by 1 mole of an ideal gas? (Use standard value.)
A) 1.0 L B) 6.02 L C) 11.2 L D) 22.4 L E) 24.0 L
40. Calculate the molar mass of CaCO_3 . (Ca = 40, C = 12, O = 16.)
A) 56 g/mol B) 68 g/mol C) 84 g/mol D) 100 g/mol E) 116 g/mol
41. Which has the lowest pH?
**A) 0.01 M HCl
B) 0.1 M NaOH
C) Pure water
D) 0.1 M CH_3COOH (acetic acid, weak)
E) 0.1 M HCl**
42. Which of the following is an aromatic compound?
A) Ethene B) Ethanol C) Benzene D) Acetylene E) Methane
43. Which functional group is present in aldehydes?
**A) $-\text{CHO}$
B) $-\text{C}=\text{O}$ (internal, no H)
C) $-\text{OH}$
D) $-\text{COOH}$
E) $-\text{NH}_2$**
44. For the equilibrium $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$, $\Delta H < 0$. Increasing temperature will:
**A) Shift the equilibrium to the right
B) Increase SO_3 yield only
C) Shift the equilibrium to the left (favours endothermic direction)
D) Have no effect on position
E) Stop the reaction**
45. The isotope ^{14}C has how many neutrons? (C atomic number = 6.)
A) 6 B) 7 C) 8 D) 14 E) 20
46. A catalyst increases the rate of a reaction by:
**A) Increasing the temperature
B) Providing an alternative pathway with lower activation energy
C) Being consumed in the net reaction
D) Increasing the activation energy
E) Shifting equilibrium toward products**
47. What type of reaction is: $\text{CH}_4 + \text{Cl}_2 \rightarrow \text{CH}_3\text{Cl} + \text{HCl}$ (in light)?
A) Addition B) Combustion C) Neutralisation D) Elimination E) Substitution

Section D - Mathematics (Q48-54)

48. Solve: $(x - 2)(x + 3) = 0$.

- A) $x = 6$
- B) $x = -6$
- C) $x = 2$ only
- D) $x = -3$ only
- E) $x = 2$ or $x = -3$

49. Solve: $x^2 - 5x + 6 = 0$.

- A) $x = 2, 3$ B) $x = 1, 5$ C) $x = -2, -3$ D) $x = 1, 6$ E) No real roots

50. Solve: $3/x = 9$. ($x \neq 0$.)

- A) $x = 27$ B) $x = 1/3$ C) $x = 1/9$ D) $x = 3$ E) $x = -3$

51. Solve the inequality: $-2x + 1 > 7$.

- A) $x > 3$ B) $x > -3$ C) $x = -3$ D) $x < -3$ E) $x < 3$

52. In a right triangle with legs 6 and 8, what is the hypotenuse?

- A) 9 B) 10 C) 11 D) 12 E) 14

53. $\sin 30^\circ = ?$

- A) $\sqrt{2}/2$ B) 0 C) 1 D) $\sqrt{3}/2$ E) $1/2$

54. A is the set $\{1, 2, 3\}$; B is the set $\{a, b\}$. How many ordered pairs are in $A \times B$ (the Cartesian product)?

- A) 2 B) 3 C) 5 D) 6 E) 9

Section E - Physics (Q55-60)

55. A body accelerates uniformly from rest at 2 m/s^2 for 5 s. What distance does it cover?

- A) 10 m B) 20 m C) 25 m D) 50 m E) 100 m

56. Two resistors, 4Ω and 6Ω , are in series across a 10 V battery. What is the current?

- A) 0.5 A B) 1 A C) 2 A D) 2.5 A E) 10 A

57. A 1.5 V cell delivers 0.3 A through a lamp. What is the lamp's resistance?

- A) 0.2Ω B) 0.45Ω C) 2Ω D) 4.5Ω E) 5Ω

58. A 2 kg block is pushed with a force of 10 N on a frictionless surface. What is its acceleration?

- A) 2 m/s^2 B) 5 m/s^2 C) 8 m/s^2 D) 10 m/s^2 E) 20 m/s^2

59. A 5 kg object is lifted 2 m. How much gravitational PE is gained? ($g = 9.8 \text{ m/s}^2$.)

- A) 9.8 J B) 10 J C) 49 J D) 98 J E) 196 J

60. The acceleration due to gravity on the Moon is about $1/6$ of that on Earth. An astronaut's weight on the Moon compared to Earth is:

- A) $1/6$ as great
- B) $1/36$ as great
- C) 6 times greater
- D) The same
- E) Zero

Answer key

This key refers to the option order printed in this paper. The online app shuffles option order on every attempt, so answer letters inside the app will differ from the ones here.

1	A	2	B	3	C	4	E	5	D
6	C	7	C	8	E	9	C	10	D
11	A	12	D	13	B	14	A	15	E
16	B	17	A	18	B	19	D	20	E
21	D	22	C	23	B	24	A	25	E
26	B	27	D	28	C	29	E	30	A
31	B	32	D	33	C	34	E	35	D
36	A	37	A	38	B	39	D	40	D
41	E	42	C	43	A	44	C	45	C
46	B	47	E	48	E	49	A	50	B
51	D	52	B	53	E	54	D	55	C
56	B	57	E	58	B	59	D	60	A